

Explainable Quantum-Inspired Multi-label Classification of Non-Functional Requirements

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Abstract. Non-functional requirements (NFRs) frequently overlap across quality attributes such as security, performance, usability, and availability, making NFR classification a natural multi-label problem. This paper presents an intrinsically explainable quantum-inspired method for multi-label NFR classification. Requirements are represented as normalized semantic states, NFR labels as projection directions, and label scores as squared projection amplitudes; a label interference matrix models NFR co-occurrence, and token-level amplitude contributions explain predictions without a post-hoc explainer. The paper also gives a theoretical characterization of the deployed decision rule, identifying when the quantum-inspired geometry reduces to cosine-style ranking and when interference, or positive-model row-wise normalization, creates genuine cross-label coupling. We evaluate quantum-inspired variants on the NICE multi-label dataset against TF-IDF Logistic Regression, TF-IDF Linear SVM, Sentence-BERT, fine-tuned DistilBERT, and a label-frequency baseline. Under 5-fold cross-validation, Sentence-BERT achieves the highest Macro-F1; the hybrid quantum-SVM model is competitive with TF-IDF baselines and obtains the highest Macro-F1 among non-embedding models, but its advantage is not statistically significant at this dataset size. The main contribution is therefore not a state-of-the-art accuracy claim, but an interpretable multi-label scoring formulation, a precise account of when its geometry helps or collapses to classical behavior, and a deletion-based faithfulness evaluation of its intrinsic token explanations.

Keywords: Requirements Engineering · Non-Functional Requirements · Multi-label Classification · Quantum-inspired Model · Explainable AI · Software Engineering

1 Introduction

Requirements Engineering (RE) is a critical activity in software development because requirement quality strongly affects downstream architecture, implementation, testing, and maintenance decisions. Among requirement types, non-functional requirements (NFRs) are particularly difficult to identify and classify automatically. NFRs describe quality attributes such as security, performance, usability, availability, scalability, and maintainability. Unlike many functional requirements, NFRs are often ambiguous, context-dependent, and overlapping.

For example, a requirement stating that payment information must be encrypted and accessible only to authorized administrators clearly relates to security. If the same requirement also constrains availability, auditability, or response time, it may express multiple NFR categories at once. Therefore, NFR classification is better treated as a multi-label problem rather than a strict single-label problem [19,21].

This paper investigates the research topic: *Explainable Quantum-Inspired Multi-label Classification of Non-Functional Requirements*. The key idea is to represent each requirement as a semantic state and estimate its degree of association with multiple NFR labels using projection amplitudes. The approach is quantum-inspired rather than quantum-hardware-based: it runs on ordinary computers and borrows mathematical intuitions such as state representation, projection amplitude, and interference from geometric information retrieval and quantum language modeling [16,18,8].

The paper’s positive claim is that this formulation provides an intrinsic explanation mechanism for multi-label NFR classification and a testable theory of when quantum-inspired geometry changes the decision rule. The basic projection component is closely related to centroid-based text classification; the interference and positive-model row-wise normalization analyses make explicit when the model becomes more than an independent centroid classifier and when it does not. Empirically, the hybrid model offers a small, suggestive accuracy gain over comparable TF-IDF baselines, while the primary value of the method lies in its transparent scoring structure and diagnostic ablation.

The contributions of this paper are:

- an intrinsically explainable quantum-inspired formulation for multi-label NFR classification;
- a theoretical characterization of interference and, for the positive-projection model, row-wise normalization as two sources of cross-label coupling;
- token-level semantic amplitude explanations evaluated with a deletion-based ERASER faithfulness test;
- empirical comparison against TF-IDF Logistic Regression, TF-IDF Linear SVM, Sentence-BERT, fine-tuned DistilBERT, and a label-frequency baseline on the NICE dataset;
- an ablation study showing where projection type, label interference, and hybrid fusion weight help or fail.

2 Related Work

Automated NFR classification includes early rule and supervised-learning work on security, performance, usability, and related quality attributes [1,7,2]. PROMISE and PROMISE-expanded support repeatable NFR experiments [11], while NICE provides the multi-label NFR setting used here [13,14]. Because label filtering, splits, and metrics differ from the NICE paper, we report all baselines on the same downloaded data.

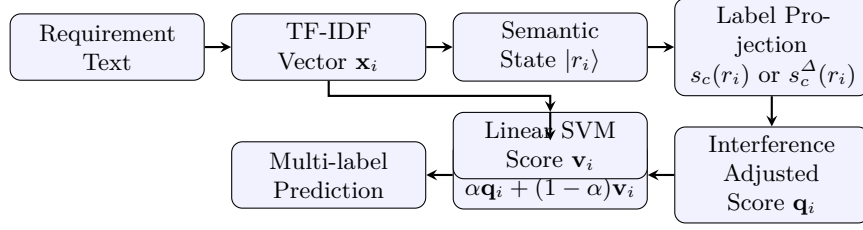


Fig. 1. Hybrid quantum-inspired multi-label classification pipeline. A requirement is transformed into a TF-IDF vector and normalized semantic state, then projected onto NFR-label directions. Co-occurrence interference adjusts the projection scores; the resulting quantum-inspired scores are fused with Linear SVM scores and thresholded to produce the multi-label prediction.

Our baselines cover TF-IDF classifiers, which remain strong in small text datasets [17], and transformer-based representations, including BERT-style transfer learning, NoBERT, and Sentence-BERT [4,6,12]. The quantum-inspired component follows geometric information retrieval and quantum-inspired language modeling, where matching is expressed through vector spaces, projections, and wave-function-inspired representations [16,18,22,8]. For explainability, we relate to LIME, SHAP, and ERASER-style rationale evaluation [15,10,5].

3 Research Questions

This study is guided by three research questions:

RQ1. Can a quantum-inspired semantic projection model classify multi-label NFRs competitively compared with classical and embedding baselines?

RQ2. Does the model provide useful explainability through token-level semantic amplitude contributions?

RQ3. Is the observed behavior stable under cross-validation rather than only in a single train/test split?

4 Proposed Method

Figure 1 summarizes the complete prediction pipeline and the point at which the quantum-inspired and classical branches are fused.

4.1 Requirement Representation

Each requirement text is first converted into a TF-IDF vector:

$$\mathbf{x}_i \in \mathbb{R}^d, \quad (1)$$

where d is the vocabulary size. The vector is normalized to form a semantic state:

$$|r_i\rangle = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2}. \quad (2)$$

This normalization allows the requirement to be interpreted as a direction in semantic space.

4.2 Label Projection

For each NFR label c , a label basis vector is estimated from the centroid of positive training examples:

$$|c\rangle = \frac{1}{N_c} \sum_{i:y_{ic}=1} |r_i\rangle, \quad (3)$$

followed by normalization. The raw association between requirement r_i and label c is computed as a squared projection amplitude:

$$s_c(r_i) = (\langle c | r_i \rangle)^2. \quad (4)$$

We call $s_c(r_i)$ an association score rather than a probability because the scores over labels are not constrained to sum to one.

4.3 Contrastive Label Projection

The positive-only centroid projection is simple and interpretable, but it does not explicitly model how a label differs from other labels. Therefore, this work also evaluates a contrastive quantum-inspired variant. For each label c , the contrastive direction is computed as:

$$|c_\Delta\rangle = \frac{\mu_c^+ - \mu_c^-}{\|\mu_c^+ - \mu_c^-\|_2}, \quad (5)$$

where μ_c^+ is the centroid of requirements that contain label c , and μ_c^- is the centroid of requirements that do not contain label c . Unlike the positive centroid vector, the contrastive direction is signed: a requirement can project in the opposite direction of a label. Squaring a negative projection would incorrectly assign a high score to evidence against the label. Therefore, the contrastive variant uses a half-wave rectified projection score:

$$s_c^\Delta(r_i) = (\max(0, \langle c_\Delta | r_i \rangle))^2. \quad (6)$$

This preserves positive alignment with the label direction while suppressing opposite-direction evidence.

4.4 Interference Adjustment

To reflect label co-occurrence, the model estimates a label interference matrix from training labels. Let N_{kl} be the number of training samples containing both labels k and l , and let N_k be the number of samples containing label k . The normalized co-occurrence matrix is:

$$M_{kl} = \frac{N_{kl}}{N_k}, \quad (7)$$

with the diagonal set to 1. The implementation mixes this matrix with the identity matrix:

$$\mathbf{M}' = (1 - \lambda)\mathbf{I} + \lambda\mathbf{M}, \quad (8)$$

where λ controls interference strength. For projection scores $\mathbf{p}_i$, adjusted scores are computed as:

$$\mathbf{s}_i = \text{normalize}(\mathbf{p}_i\mathbf{M}'). \quad (9)$$

For the positive-projection model, $\text{normalize}(\cdot)$ denotes row-wise min-max normalization into $[0, 1]$. For the contrastive model, the rectified projection is passed through a sigmoid calibration and the interference-adjusted scores are clipped into $[0, 1]$. The resulting scores are association scores for thresholding rather than mutually exclusive probabilities.

4.5 Hybrid Quantum-SVM Fusion

To improve stability on small and imbalanced datasets, a hybrid model combines the contrastive quantum-inspired score with a one-vs-rest Linear SVM score:

$$\mathbf{h}_i = \alpha\mathbf{q}_i + (1 - \alpha)\mathbf{v}_i, \quad (10)$$

where $\mathbf{q}_i$ is the contrastive quantum-inspired score vector, $\mathbf{v}_i$ is the calibrated SVM score vector, and α is the quantum fusion weight. In the final experiments, $\alpha = 0.30$ is used as a moderate quantum weight; the ablation study (Table 5) shows that Macro-F1 is not sensitive to the exact value in this range ($\alpha = 0.15$, 0.30 , and 0.50 are within one standard deviation of each other), so $\alpha = 0.30$ was kept as a round, non-edge value rather than as a value selected for maximizing cross-validation Macro-F1.

4.6 Explainability

For a predicted label c , token-level contribution is estimated from the element-wise product between the requirement state and the label basis:

$$e_{ijc} = r_{ij}c_j. \quad (11)$$

Terms with the largest absolute contributions are reported as explanations. In the experiments, these explanations are generated by the rectified contrastive projection model. We evaluate them using a deletion test: if the highlighted terms are faithful, deleting them should reduce the model score more than deleting random nonzero terms.

4.7 Theoretical Characterization

The purpose of this analysis is boundary characterization rather than a claim that every quantum-inspired mechanism improves the recommended hybrid model. It identifies which operations create behavior beyond independent centroid scoring and which collapse to classical geometry. The positive-projection formulation exposes two mechanistically distinct sources of cross-label coupling in its deployed decision rule: the interference matrix (Proposition 1) and the row-wise score normalization applied by this variant before thresholding (Proposition 2). Only the within-instance *ranking* of labels reduces to a cosine-similarity ranking; the thresholded binary decision of this variant does not reduce to L independent per-label centroid classifiers, even when interference is switched off. The contrastive and hybrid variants instead use sigmoid-calibrated scores without row-wise min-max normalization, so Proposition 2 does not apply to them. For the recommended hybrid, Proposition 1 characterizes its quantum branch, but the ablation later shows that this coupling is weak or harmful on NICE; its theoretical value is therefore to delimit a mechanism, not to present that mechanism as the source of the hybrid’s empirical result.

Proposition 1 (Interference as label coupling). *With $\mathbf{M}' = (1 - \lambda)\mathbf{I} + \lambda\mathbf{M}$, the pre-normalization score for label l has the form*

$$u_l \propto (1 - \lambda)p_l + \lambda \sum_k p_k M_{kl}. \quad (12)$$

The second term injects score mass from co-occurring labels k , where $M_{kl} = N_{kl}/N_k$. This coupling cannot be represented by a diagonal per-label centroid classifier; together with the row-wise normalization in Proposition 2, it is one of two mechanisms by which the positive-projection decision rule departs from an independent per-label classifier.

Proof. Expanding $\mathbf{pM}'$ gives $\mathbf{p}((1 - \lambda)\mathbf{I} + \lambda\mathbf{M})$, so the l -th component is $(1 - \lambda)p_l + \lambda \sum_k p_k M_{kl}$. If $\mathbf{M}'$ were diagonal, each adjusted label score would depend only on its own projection score p_l . The off-diagonal entries M_{kl} for $k \neq l$ therefore introduce explicit cross-label coupling through co-occurrence statistics.

Proposition 2 (Ranking invariance under row-wise normalization). *Let $\mathbf{u}_i = \mathbf{p}_i\mathbf{M}'$ be the pre-normalization score vector from Proposition 1, and let $\mathbf{s}_i = \text{normalize}(\mathbf{u}_i)$ (row-wise min-max into $[0, 1]$) be the score vector actually thresholded by the implementation. For any two labels k, l evaluated on the same instance i ,*

$$s_{i,k} \geq s_{i,l} \iff u_{i,k} \geq u_{i,l};$$

in particular, at $\lambda = 0$ (so $u_{i,l} = p_{i,l} = \langle l | r_i \rangle^2$), this ranking equals the ranking by cosine similarity $\langle l | r_i \rangle$. However, for a fixed global threshold $\tau \in (0, 1)$, the decision $s_{i,l} \geq \tau$ is equivalent to

$$u_{i,l} \geq \min_k u_{i,k} + \tau \left(\max_k u_{i,k} - \min_k u_{i,k} \right) =: \tau_i,$$

where τ_i depends on instance i through the scores of every label on that instance, not only label l . Consequently, τ does not correspond to a single instance-independent cutoff on $u_{i,l}$ (or, at $\lambda = 0$, on cosine similarity): two instances with an identical raw score for label l can receive opposite decisions for that label if their scores on other labels differ.

Proof. For fixed i with $\max_k u_{i,k} > \min_k u_{i,k}$, the normalized score is

$$s_{i,l} = \frac{u_{i,l} - \min_k u_{i,k}}{\max_k u_{i,k} - \min_k u_{i,k}}. \quad (13)$$

It is a strictly increasing affine function of $u_{i,l}$ with the same slope and intercept for every label l of instance i , which proves the ranking-invariance claim. At $\lambda = 0$, $u_{i,l} = \langle l \mid r_i \rangle^2$, and since $\langle l \mid r_i \rangle = \cos \theta_{r_i,l} \geq 0$ (TF-IDF features and positive centroids are non-negative), squaring is order-preserving on $[0, \infty)$, so the ranking by $u_{i,l}$ equals the ranking by $\cos \theta_{r_i,l}$. Solving $s_{i,l} \geq \tau$ for $u_{i,l}$ using the definition of $s_{i,l}$ gives the threshold τ_i above, which is a function of $\min_k u_{i,k}$ and $\max_k u_{i,k}$ and therefore of the scores of labels $k \neq l$ on instance i ; two instances with equal $u_{i,l}$ but different label-score profiles can therefore yield different values of τ_i relative to $u_{i,l}$, hence different decisions. When $\max_k u_{i,k} = \min_k u_{i,k}$, the implementation sets $s_{i,l} = 0$ for all l , so no label is predicted for instance i at any $\tau > 0$.

Remark 1. For two labels and $\tau = 0.5$, $\mathbf{u} = (0.5, 0.0)$ predicts label 1 after normalization, whereas $\mathbf{u} = (0.5, 0.9)$ gives the same raw label-1 score but does not predict label 1. Row-wise normalization is therefore a second source of cross-label coupling, independent of $\mathbf{M}'$.

5 Experimental Setup

5.1 Datasets

NICE multi-label dataset. The main experiment uses the NICE dataset from Zenodo [13]. The downloaded CSV contains 622 requirements and binary label columns for NFR classes. After filtering rows with at least one NFR sub-class label and removing the empty “Other” label, the experiment uses 381 requirements and 11 NFR labels. Among these, 125 requirements contain more than one NFR label, giving a label cardinality of 1.3648.

PROMISE-expanded dataset. A secondary experiment uses PROMISE-expanded from the public requirements classification repository [11]. This dataset is treated as an 11-class single-label NFR subtype classification task and is used as an auxiliary sanity check rather than the main multi-label experiment.

5.2 Compared Models

The compared baselines are:

- **Label-frequency baseline:** predicts labels according to training label frequencies.
- **TF-IDF Logistic Regression:** one-vs-rest Logistic Regression using unigram and bigram TF-IDF features.
- **TF-IDF Linear SVM:** one-vs-rest Linear SVM using unigram and bigram TF-IDF features.
- **Sentence-BERT + LR:** all-MiniLM-L6-v2 embeddings followed by one-vs-rest Logistic Regression.
- **Fine-tuned DistilBERT:** distilbert-base-uncased fine-tuned directly for multi-label classification with BCE loss and per-label threshold calibration.

The quantum-inspired models are:

- **Quantum-inspired Projection:** positive-centroid semantic projection with label interference.
- **Contrastive Quantum Projection:** positive-minus-negative semantic projection with label interference.
- **Hybrid Quantum-SVM Fusion:** score-level fusion of contrastive quantum projection and Linear SVM, using a quantum fusion weight of 0.30.

5.3 Metrics and Threshold Selection

The main reported metrics are Micro-F1, Macro-F1, Hamming loss, and Label Ranking Average Precision (LRAP). Macro-F1 is emphasized because the NFR labels are imbalanced. Two threshold protocols are evaluated. First, a single global threshold is selected on an internal validation split by maximizing Macro-F1 over thresholds from 0.05 to 0.95. Second, a per-label threshold protocol selects an independent threshold for each NFR label on the validation split. Fine-tuned DistilBERT was evaluated only with the per-label protocol; no global-threshold DistilBERT result is reported.

5.4 Hyperparameters

Table 1 summarizes the main hyperparameters used for reproducibility.

6 Results

6.1 Single Train/Validation/Test Split on NICE

Table 2 reports results from a single train/validation/test split on the NICE dataset.

On this single split, Sentence-BERT obtains the highest Macro-F1 and LRAP. The positive projection variant performs better than the rectified contrastive variant, while the hybrid quantum-SVM model and the contrastive quantum projection share the lowest Hamming loss on this split. Because the split contains only 381 usable instances, these single-split results should be interpreted cautiously.

Table 1. Main hyperparameters used in the experiments.

Component	Setting
TF-IDF features	unigrams+bigrams, min_df=1
Sublinear TF-IDF	enabled for contrastive, hybrid, and SVM-only ablation
Logistic Regression	one-vs-rest, class_weight=balanced, max_iter=3000
Linear SVM	one-vs-rest LinearSVC, class_weight=balanced
Sentence-BERT	all-MiniLM-L6-v2 + one-vs-rest Logistic Regression
Fine-tuned DistilBERT	2 epochs, batch size 16, max length 128, learning rate 5×10^{-5}
Positive projection interference	$\lambda = 0.15$
Contrastive projection interference	$\lambda = 0.05$
Hybrid projection interference	$\lambda = 0.03$
Hybrid quantum weight	$\alpha = 0.30$ unless varied in ablation
Threshold grid	0.05 to 0.95 with step 0.05
Random seed	42

Table 2. NICE multi-label results on a single train/validation/test split, comparing label-frequency, TF-IDF, Sentence-BERT, and quantum-inspired models. “Pos.”, “Cont.”, and “Q-SVM” abbreviate positive projection, contrastive projection, and hybrid quantum-SVM.

Model	Threshold	Micro-F1	Macro-F1	Hamming Loss	LRAP
Label-frequency	0.05	0.2221	0.2166	0.8751	0.4183
TF-IDF LR	0.45	0.6299	0.5266	0.0901	0.7861
TF-IDF Linear SVM	0.45	0.6212	0.5205	0.0877	0.7843
Sentence-BERT + LR	0.50	0.6667	0.6049	0.0901	0.8286
Pos. projection	0.85	0.6076	0.5618	0.0980	0.7694
Cont. projection	0.50	0.4907	0.3885	0.0870	0.6818
Hybrid Q-SVM	0.45	0.6284	0.5286	0.0870	0.7779

6.2 5-Fold Cross-validation on NICE

Table 3 reports 5-fold cross-validation results with mean and standard deviation across folds under the global-threshold protocol.

Table 3. Mean and standard deviation of 5-fold NICE results using one validation-selected global threshold per fold. “Pos.”, “Cont.”, and “Q-SVM” abbreviate positive projection, contrastive projection, and hybrid quantum-SVM.

Model	Micro-F1	Macro-F1	Hamming Loss	LRAP
Sentence-BERT + LR	0.6493 $\pm$ 0.0357	0.6121 $\pm$ 0.0450	0.0919 $\pm$ 0.0200	0.8361 $\pm$ 0.0225
Hybrid Q-SVM	0.6839 $\pm$ 0.0331	0.6084 $\pm$ 0.0294	0.0768 $\pm$ 0.0074	0.8040 $\pm$ 0.0266
TF-IDF Linear SVM	0.6821 $\pm$ 0.0405	0.6049 $\pm$ 0.0336	0.0764 $\pm$ 0.0094	0.8056 $\pm$ 0.0262
TF-IDF LR	0.6787 $\pm$ 0.0463	0.5977 $\pm$ 0.0229	0.0759 $\pm$ 0.0140	0.7990 $\pm$ 0.0240
Cont. projection	0.4300 $\pm$ 0.0783	0.4118 $\pm$ 0.0275	0.2074 $\pm$ 0.1634	0.6968 $\pm$ 0.0503
Pos. projection	0.5931 $\pm$ 0.0166	0.5490 $\pm$ 0.0237	0.1096 $\pm$ 0.0213	0.7830 $\pm$ 0.0304
Label-frequency	0.2328 $\pm$ 0.0034	0.2083 $\pm$ 0.0032	0.7927 $\pm$ 0.0019	0.4059 $\pm$ 0.0110

Under standard 5-fold cross-validation with a global threshold, Sentence-BERT is the strongest Macro-F1 and LRAP baseline. The hybrid quantum-SVM fusion model is the strongest non-embedding model in Micro-F1 and Macro-F1 and improves over both pure quantum-inspired variants. A Wilcoxon signed-rank test over fold-level Macro-F1 finds no significant difference between the hybrid model and TF-IDF Linear SVM ($p = 0.1875$) or TF-IDF Logistic Regression ($p = 0.4375$). Given the small number of folds and the dependence induced by cross-validation resampling, these tests should be interpreted as descriptive evidence rather than a strong statistical conclusion [3].

6.3 5-Fold Cross-validation with Per-label Thresholds

Table 4 reports the 5-fold results when each label receives its own threshold selected on the validation split.

Table 4. Mean and standard deviation of 5-fold NICE results with per-label threshold calibration. This is the only protocol run for fine-tuned DistilBERT; “Q-SVM” abbreviates hybrid quantum-SVM.

Model	Micro-F1	Macro-F1	Hamming Loss	LRAP
Sentence-BERT + LR	0.6386 $\pm$ 0.0280	0.5990 $\pm$ 0.0276	0.0947 $\pm$ 0.0092	0.8361 $\pm$ 0.0225
Hybrid Q-SVM	0.6178 $\pm$ 0.0749	0.5854 $\pm$ 0.0341	0.1027 $\pm$ 0.0476	0.8040 $\pm$ 0.0266
TF-IDF Linear SVM	0.6085 $\pm$ 0.0588	0.5640 $\pm$ 0.0309	0.1057 $\pm$ 0.0378	0.8056 $\pm$ 0.0262
TF-IDF LR	0.6219 $\pm$ 0.0588	0.5629 $\pm$ 0.0532	0.0967 $\pm$ 0.0195	0.7990 $\pm$ 0.0240
Fine-tuned DistilBERT	0.5420 $\pm$ 0.0569	0.5333 $\pm$ 0.0591	0.1402 $\pm$ 0.0361	0.7105 $\pm$ 0.0516

With per-label threshold calibration, Sentence-BERT again obtains the best Macro-F1, and the hybrid quantum-SVM model remains the strongest non-embedding model in Macro-F1. Macro-F1 decreases from the global-threshold

protocol (Table 3), plausibly because each label has fewer validation examples. DistilBERT was run only under this protocol and is weaker than SentenceBERT+LR and the TF-IDF/hybrid models at the present data scale.

6.4 Ablation Study

Table 5 reports a 5-fold ablation study of the main quantum-inspired components. The ablation compares the original positive projection, contrastive projection with and without interference, an SVM-only component using the same sublinear TF-IDF representation as the hybrid model, and hybrid fusion with different quantum weights.

Table 5. Ablation study on the NICE multi-label dataset. Hybrid fusion with $\alpha \in \{0.15, 0.30, 0.50\}$ performs within one standard deviation across all four metrics, indicating that the projection score is useful as a calibrated auxiliary signal but that Macro-F1 is not sensitive to the exact quantum weight in this range; $\alpha = 0.30$ is reported as the main configuration.

Variant	Micro-F1	Macro-F1	Hamming Loss	LRAP
Hybrid $\alpha = 0.15$	0.6860 $\pm$ 0.0312	0.6091 $\pm$ 0.0293	0.0756 $\pm$ 0.0072	0.8085 $\pm$ 0.0280
Hybrid $\alpha = 0.30$	0.6839 $\pm$ 0.0331	0.6084 $\pm$ 0.0294	0.0768 $\pm$ 0.0074	0.8040 $\pm$ 0.0266
Hybrid $\alpha = 0.50$	0.6814 $\pm$ 0.0404	0.6054 $\pm$ 0.0299	0.0795 $\pm$ 0.0113	0.8012 $\pm$ 0.0232
SVM-only	0.6713 $\pm$ 0.0553	0.5945 $\pm$ 0.0409	0.0843 $\pm$ 0.0263	0.8081 $\pm$ 0.0291
Pos. proj. w/o interf.	0.5948 $\pm$ 0.0238	0.5526 $\pm$ 0.0285	0.1108 $\pm$ 0.0233	0.7839 $\pm$ 0.0309
Pos. proj. w/ interf.	0.5931 $\pm$ 0.0166	0.5490 $\pm$ 0.0237	0.1096 $\pm$ 0.0213	0.7830 $\pm$ 0.0304
Cont. proj. w/o interf.	0.5447 $\pm$ 0.0420	0.5397 $\pm$ 0.0398	0.1339 $\pm$ 0.0100	0.7183 $\pm$ 0.0523
Cont. proj. w/ interf.	0.4300 $\pm$ 0.0783	0.4118 $\pm$ 0.0275	0.2074 $\pm$ 0.1634	0.6968 $\pm$ 0.0503

Interference is neutral for positive projection (Macro-F1 0.5526 without vs. 0.5490 with it) and harmful for contrastive projection (0.5397 vs. 0.4118), so it is a diagnostic mechanism rather than an accuracy contribution here. The rectified contrastive formulation remains semantically safer than squaring negative evidence but is weaker than positive projection. Hybrid $\alpha = 0.30$ raises the SVM-only Macro-F1 from 0.5945 to 0.6084; thus the projection score is most useful as an auxiliary signal, not the dominant one.

6.5 Explainability Evaluation

Across 158 true test-label assignments, deleting the contrastive model’s top three terms gives comprehensiveness 0.0305, 5.21 times random deletion (0.0059). Sufficiency is $f(x)_c - f(x_{\text{top-3}})_c$, where lower is better; its value is -0.0395 versus 0.0337 for random terms. The negative value means the rationale alone scores slightly higher than the full input, possibly because other tokens add noise; it is not evidence of human-rationale quality. The SVM coefficient baseline obtains comprehensiveness 0.0867 (4.01 times random) and sufficiency 0.0484 (0.42 times random). Thus both are faithful relative to their controls, while SVM removes

more absolute score mass; the projection explanation is not claimed to be uniformly stronger.

Table 6 gives a qualitative example from the same deletion test. The highest-contribution terms for the Security label correspond to access control concepts that are also plausible human rationales.

Table 6. Example requirement with top token-level contributions for the true NFR label, illustrating the intrinsic explanation produced by the projection model.

Requirement excerpt	Label	Top terms
“The product shall ensure that it can only be accessed by authorized users”	Security	authorized, only, access

6.6 Auxiliary PROMISE-expanded Result

On one 70/30 PROMISE-expanded split, projection obtains 0.7025 accuracy and 0.6748 Macro-F1 versus 0.6899 and 0.6344 for TF-IDF Logistic Regression. Without cross-validation or a significance test, this single-label result is only a sanity check and does not test multi-label interference.

7 Discussion

Sentence-BERT leads Macro-F1 and LRAP; hybrid quantum-SVM remains competitive with TF-IDF baselines and provides intrinsic token contributions. Pure projections lose to Linear SVM in all five folds ($d_z = -1.36$ positive and -5.92 contrastive; Wilcoxon $p = 0.0625$, the minimum at $n = 5$).

Hybrid improves over the matched SVM-only baseline by 0.0139 Macro-F1 ($d_z = 0.49$), but loses in two folds and all main Wilcoxon tests are non-significant ($p \geq 0.1875$). An idealized paired test would require 34 independent pairs for 80% power; dependent cross-validation folds cannot supply them. The gain is therefore suggestive, not confirmed.

A paired test-set bootstrap agrees: hybrid minus Linear SVM is 0.0081 (95% CI $[-0.0100, 0.0263]$) and hybrid minus Logistic Regression is 0.0020 ($[-0.0205, 0.0227]$). Only hybrid versus pure projection is clearly separated ($+0.140$, $[0.074, 0.213]$, $p < 0.001$), showing that the projection depends on discriminative calibration to be practically useful.

Unlike Sentence-BERT’s dense embeddings, the projection model exposes token-level contributions without post-hoc LIME [15] or SHAP [10]. It is appropriate when intrinsic interpretability and a small dependency footprint matter more than maximum Macro-F1, not as a confirmed accuracy improvement.

8 Threats to Validity

Dataset size and availability. NICE contains 381 usable multi-label NFR requirements after filtering. Results may depend on its vocabulary, label distribution, and annotation, so a second multi-label corpus is needed for broad generalization. PROMISE-expanded is only a single-label sanity check and does not test interference; no state-of-the-art claim is made beyond NICE.

Baseline scope. The current implementation includes TF-IDF baselines, a frozen Sentence-BERT encoder, and fine-tuned DistilBERT, but not NoRBERT, RoBERTa, ML-kNN, or the NICE small-language model. Other encoders and schedules could change the ranking; the central claim concerns interpretability and diagnostic ablation, not state-of-the-art accuracy.

Model simplicity. The quantum-inspired models are based on TF-IDF semantic states, projection directions, and score fusion. They do not yet use contextual neural embeddings inside the projection component.

Threshold sensitivity. Multi-label classification depends strongly on decision thresholds. The experiments show that per-label threshold calibration can change the relative ranking of models.

Explainability evaluation. The deletion test measures faithfulness to the model’s own scoring function, but it does not measure human usefulness or agreement with expert rationales.

Ablation scope. The ablation study examines projection type, interference, and fusion weight, but it does not yet test contextual embeddings inside the quantum-inspired projection component.

9 Conclusion

This paper presented an intrinsically explainable quantum-inspired formulation for multi-label NFR classification. Requirements are semantic states, labels are projection directions, and predictions are explained through semantic amplitude contributions rather than post-hoc approximations. The theoretical analysis clarifies exactly when the model behaves like cosine-style centroid ranking and when interference, or row-wise normalization in the positive-projection variant, creates cross-label coupling.

Empirically, the method’s strongest role is explanatory and diagnostic. The hybrid quantum-SVM model is competitive with TF-IDF baselines and is the strongest non-embedding model in Macro-F1, but its accuracy advantage is not statistically significant at the current dataset size; pure projection variants are weaker as standalone classifiers. The deletion-based evaluation shows that top contribution terms are substantially more influential than random terms, supporting the faithfulness of the intrinsic explanations. Thus, the paper contributes

a transparent multi-label scoring framework and a careful account of when its quantum-inspired geometry helps, not an inflated accuracy claim.

Future work should integrate contextual embeddings, compare with broader BERT/RoBERTa [9], NoRBERT, and ML-kNN [20] baselines, test a second multi-label NFR corpus, and collect human rationale judgments.

Reproducibility

The artifact includes all scripts, data-processing code, reports, and pinned dependencies needed to reproduce the experiments, including an aggregate runner script. For anonymous review, it is available through the anonymous artifact repository.

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